MIL-C-50344A(MU)
15 April 1971
SUPERSEDING
MIL-C-50344(MU)
10 November 1969

MILITARY SPECIFICATION

COUNTER, ELECTRONIC: 10548009

1. SCOPE

1.1 This specification covers a pulse repetition frequency counter (PRF counter) used in Subassembly Test Set MX-8048/TPM-22, part of Radar Test Set AN/TPM-22, and Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10548009	PRF Counter, Electronic

PACKAGING DATA SHEET

10548009	Packaging of PRF Counter, Electronic

(Copies of specifications, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The PRF counter shall be manufactured in accordance with Drawing F10548009 and drawings pertaining thereto.

3.2 General specifications. - Where applicable, the PRF counter shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

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3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The PRF counter shall be mounted on the vibration table in normal service position. The PRF counter shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The PRF counter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. - The PRF counter shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The PRF counter shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The PRF counter shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

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3.3.4.2 Nonoperating (storage). - The PRF counter shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65° F and +160° F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). - The PRF counter shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the PRF counter.

3.5 Performance. - When supplied with the inputs specified in Table I, the PRF counter shall yield the outputs specified in Table I.

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TABLE I. - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTICS	TERMINALS
INPUT POWER	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak capacity 300 mA.	P1-30, P1-31, & P1-32 (+5 V); P1-24, P1-25, & P1-26 (RET)
INPUT SIGNALS:		
Cp	7.5 MHz square wave or sine wave, amplitude 4.3 to 5.2 V peak-to-peak.	P1-47
Cpd	Same as Cp, except delayed 20 to 40 ns from Cp.	P1-45
OUTPUT SIGNALS:		
$\overline{\text{PRF RESET}}$	20-kHz negative pulse (from not less than +3 Vdc level to zero V and return), 0.067 μs wide.	P1-52 & P1-53
$\overline{\text{PRF CAL}}$	20-kHz negative pulse pair (from not less than +3 Vdc level to zero V and return), displaced 127 clock pulses (16.89 μs) from $\overline{\text{PRF RESET}}$.	P1-37
$\overline{\text{STROBE}}$	Train of 15 negative pulses (from not less than +3 Vdc level to zero V and return), 0.133 μs wide, on 1.064 μs centers, leading edge of first pulse displaced 261 clock pulses (34.713 μs) from leading edge of $\overline{\text{PRF RESET}}$.	P1-12

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TABLE I. - INPUT AND OUTPUT CHARACTERISTICS (Cont)

REQUIREMENT	CHARACTERISTIC	TERMINALS
$\overline{T8}$	Same as $\overline{T1}$, except displaced 314 clock pulses (41.762 μs) from $\overline{PRF\ RESET}$.	P1-1
$\overline{T9}$	Same as $\overline{T1}$, except displaced 322 clock pulses (42.826 μs) from $\overline{PRF\ RESET}$.	P1-1
$\overline{T10}$	Same as $\overline{T1}$, except displaced 330 clock pulses (43.890 μs) from $\overline{PRF\ RESET}$.	P1-9
$\overline{T11}$	Same as $\overline{T1}$, except displaced 338 clock pulses (44.954 μs) from $\overline{PRF\ RESET}$.	P1-4
$\overline{PRF}$	1) When P1-53 is connected to P1-41, 20-kHz negative pulse (from not less than +3 Vdc level to zero V and return), approximately 0.6 μs wide, leading edge delayed approximately 20 ns from leading edge of $\overline{PRF\ RESET}$.	P1-10

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TABLE I. - INPUT AND OUTPUT CHARACTERISTICS (Cont)

REQUIREMENT	CHARACTERISTICS	TERMINALS
$\overline{\text{PRF}}$	2) When P1-37 is connected to P1-41, 20-kHz negative pulse (from not less than +3 Vdc level to zero V and return), approximately 0.73 μs wide, leading edge delayed 127 clock pulses (approximately 16.89 μs) from leading edge of $\overline{\text{PRF}}$ RESET.	

NOTE: The delays specified in this table may be slightly longer because of propagation delays. Frequencies and time delays specified will vary with changes of the input frequency.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three PRF counters manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3

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and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in Table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60° F and 90° F).

TABLE II. - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One PRF counter shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III. Three PRF counters shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

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TABLE III. - SPECIAL SAMPLING

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings-Visual
302. General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE IV. - SPECIAL SAMPLING, ENVIRONMENTAL

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude	3.3.3	4.6.1.3
306. Temperature	3.3.4	4.6.1.4
307. Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable TDP.

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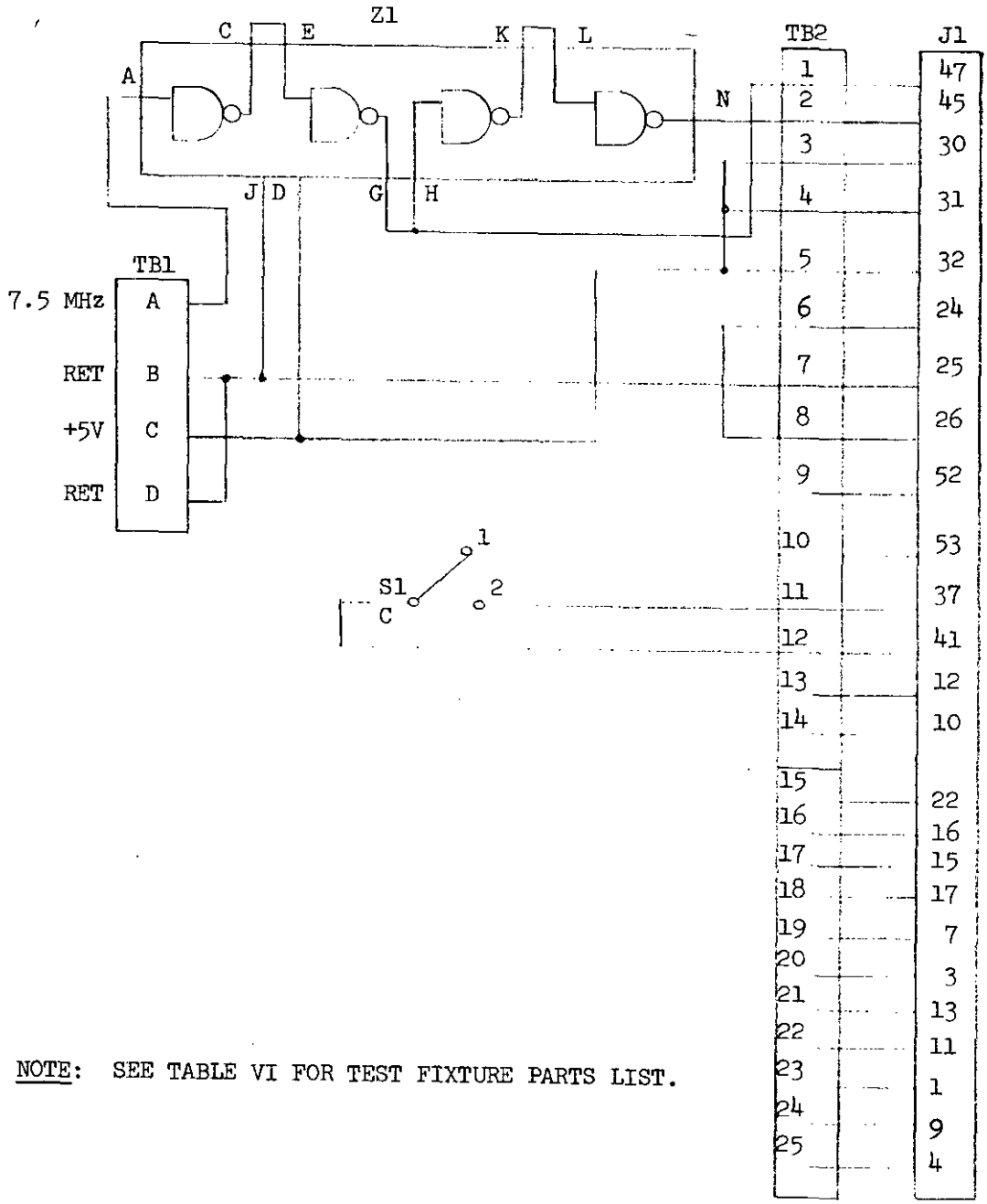
4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. - In conjunction with 4.5.2, the standard test equipment listed in Table V shall be utilized in performing the applicable test of 4.6.

TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2. Signal generator	7.5 MHz square wave or sine wave amplitude 5.0 ±0.2 V peak-to-peak, into 50-ohm load.
3. Test fixture	As specified in Figure 1. See Table VI for test fixture parts list.
4. Oscilloscope	Dual channel rise time of scope plus measuring probe 10 ns or less. Sweep period 50 ns per division or greater. Input resistance greater than 8 megohms. Input capacitance less than 25 μF.

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NOTE: SEE TABLE VI FOR TEST FIXTURE PARTS LIST.

FIGURE 1. TEST FIXTURE SETUP.

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TABLE VI. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
TB1	Terminal board, 37TB-4 conforming to MIL-T-55164/1 or equivalent.
TB2	Terminal board, 37TB-4 conforming to MIL-T-55164/1 or equivalent.
TB3	Terminal board 37TB-12 conforming to MIL-T-55164/1 or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-1, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
Z1	Integrated Circuit, SG-140-03, Sylvania or equivalent.
S1	Switch, SPDT.

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD-810 except that failure criteria do not apply. After vibration, the PRF counter shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

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4.6.1.2 Shock. - This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the PRF counter shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 in Hg). Operation of Step 3 shall be the performance tests specified in 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160° F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155° F. In Step 5, the temperature shall be maintained for 4 hours and the PRF counter shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

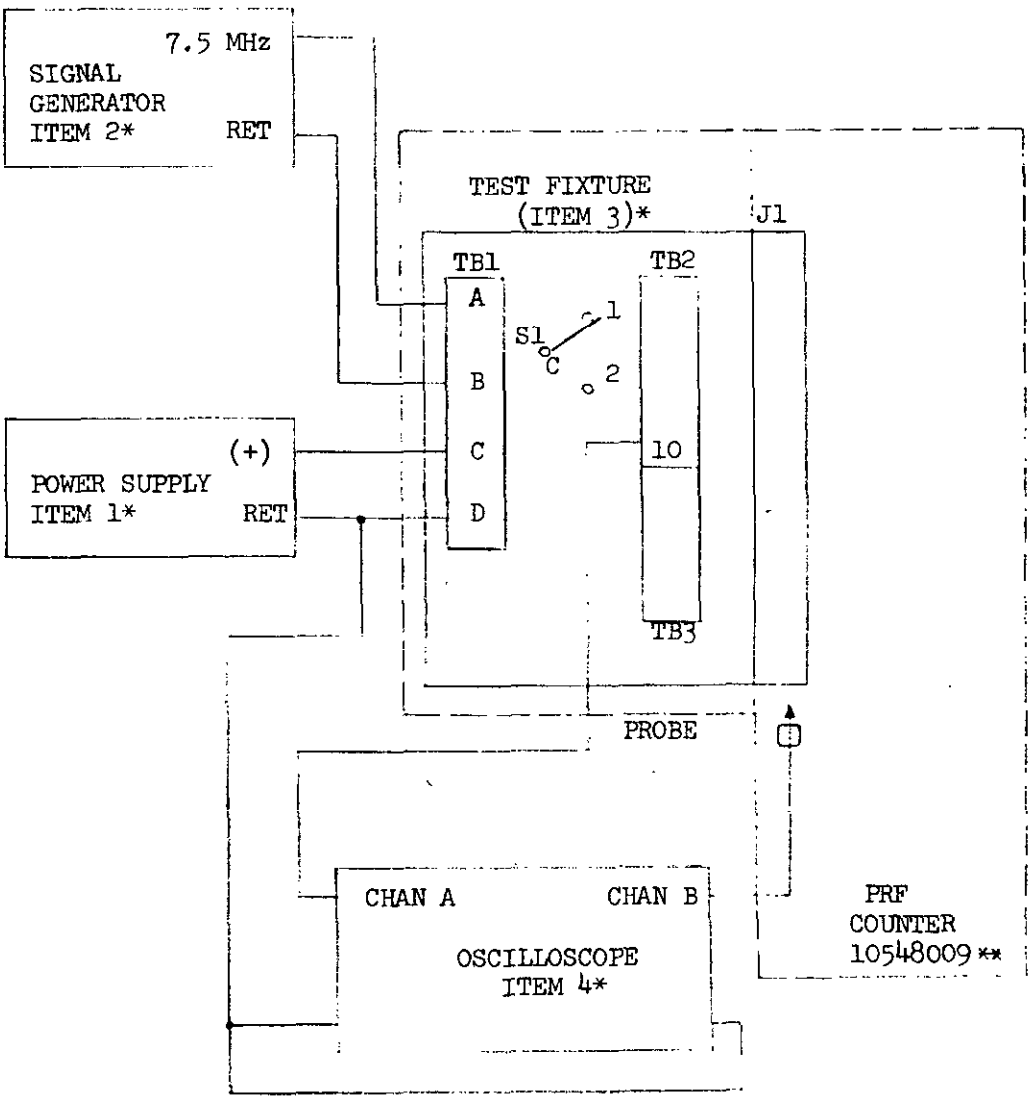
4.6.1.4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65° F maintained for not less than 4 hours. The temperature of Step 4 shall be -40° F. In Steps 5 and 7, the PRF counter shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating). - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The PRF counter shall be tested in accordance with 4.6.3.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - Adjust power supply and signal generator, items 1 and 2 of Figure 2, to deliver outputs described as "Input power" and "Input signal (Cp)" in Table I. Connect test equipment, items 1 through 4, as shown in Figure 2. Insert PRF counter into jack J1 in test fixture, item 3. Output signal labeled $\overline{\text{PRF RESET}}$ in Table I shall appear on channel A of oscilloscope, item 4. Use channel B of oscilloscope to check for presence of output signals of Table I at the terminals of test fixture, item 3 shown in Table VII.

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*REFER TO 4.5.3 -

** REFER TO 1.1

FIGURE 2. PRF COUNTER TEST SETUP.

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TABLE VII. - TEST FIXTURE OUTPUT SIGNAL TERMINALS

OUTPUT SIGNAL	TEST FIXTURE TERMINAL
$\overline{\text{PRF RESET}}$	TB2-9 & TB2-10
$\overline{\text{PRF CAL}}$	TB2-11
$\overline{\text{STROBE}}$	TB2-13
$\overline{\text{T1}}$	TB3-15
$\overline{\text{T2}}$	TB3-16
$\overline{\text{T3}}$	TB3-17
T4	TB3-18
$\overline{\text{T5}}$	TB3-19
$\overline{\text{T6}}$	TB3-20
$\overline{\text{T7}}$	TB3-21
$\overline{\text{T8}}$	TB3-22
$\overline{\text{T9}}$	TB3-23
$\overline{\text{T10}}$	TB3-24
$\overline{\text{T11}}$	TB3-25
$\overline{\text{PRF}}$, condition 1	TB2-14, with switch S1 in pos. 1
$\overline{\text{PRF}}$, condition 2	TB2-14, with switch S1 in pos. 2

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548009. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. - The PRF counter is used in Subassembly Test Set MX-8048/TPM-22, part of Radar Test Set AN/TPM-22, and in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The PRF counter is a digital counter. It counts down from a 7.5 megahertz (MHz) input signal and yields output signals of specified duration at fixed intervals. In the radar set these output signals are used to generate the radar PRF and timing pulses while in the test set they are used in checking the performance of the receiver of Radar Set AN/VPS-2.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A235

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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